



Solidification Intensification in Dual-Cooled Thermal Energy Storage Using Porous Metal Foam and Ternary Nanofluid

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Abstract: Efficient solidification and rapid heat extraction are essential for improving the charging and discharging performance of cold thermal energy storage systems. However, the low thermal conductivity of conventional phase-change media continues to limit their thermal response. This study investigates the combined effects of dual cooling boundaries, porous metal foam, and a ternary nanofluid on the solidification performance of a cold thermal energy storage system with a complex enclosure geometry. A transient numerical model was developed using the Galerkin finite element method with implicit time integration. The ternary nanofluid, consisting of Al_2O_3 , TiO_2 , and Ag nanoparticles dispersed in water, was represented using a homogeneous mixture model, while fluid motion was neglected under the solidification conditions considered. The effects of nanoparticle loading and metal foam were evaluated in terms of temperature distribution, solid-fraction evolution, solidification-front propagation, and freezing time. The results showed that the ternary nanofluid reduced the solidification time by approximately 13.11%, whereas the introduction of metal foam produced a substantially larger reduction of about 82.22%. The combined configuration reduced the freezing time from 322.62 s for the baseline case to 49.84 s, corresponding to an overall reduction of approximately 84.55%. The metal foam exerted the dominant influence by establishing continuous high-conductivity pathways throughout the storage medium, while the ternary nanoparticles provided an additional improvement in thermal transport. These findings demonstrate that coupling porous conductive structures with multicomponent nanofluids is an effective strategy for accelerating solidification and provide useful guidance for the thermal design of compact and high-response cold energy storage systems.

Keywords: Cold thermal energy storage; Phase change material; Solidification; Porous metal foam; Ternary nanofluid; Heat transfer intensification

1 Introduction

Improving heat transfer during phase change is an important requirement for the efficient operation of thermal energy storage systems. Porous metal foams have attracted considerable attention in this context because their interconnected structure combines a large specific surface area with high thermal conductivity. When incorporated into thermal systems, the solid matrix provides continuous heat-transfer pathways and promotes more effective thermal transport, making metal foams suitable for heat exchangers, cooling devices, and thermal energy storage units [1, 2]. Another approach to improving thermal transport is the use of multicomponent nanofluids. In particular, ternary nanofluids containing three types of nanoparticles have received increasing attention because their effective thermophysical properties can differ from those of conventional fluids and nanofluids containing only one or two nanoparticle species. Appropriate combinations of nanoparticles can improve thermal conductivity and heat-transfer performance and have consequently been investigated in cooling, thermal management, and energy storage applications [3, 4].

Phase change materials (PCMs) are widely employed in thermal energy storage because they can store and release substantial amounts of thermal energy over a relatively narrow temperature range. Their practical performance, however, is often constrained by low thermal conductivity, which slows heat transfer and extends the time required

for charging and discharging. Nanoparticle-enhanced PCMs (NEPCMs) have therefore been investigated as a means of modifying the effective thermal properties of the storage medium and accelerating phase-change processes [5, 6]. Porous metal foams provide a complementary mechanism. Rather than modifying only the properties of the fluid or PCM, the interconnected metallic skeleton establishes conductive pathways throughout the storage domain and can substantially shorten melting or solidification times. Combining porous structures with nanoparticle-based thermal enhancement therefore offers a promising route for addressing the heat-transfer limitations of conventional PCM-based storage systems and improving their applicability in energy-efficient thermal management and renewable-energy systems [7, 8]. The relevance of PCM-based thermal storage to cooling applications was also highlighted by Bista et al. [9], who reviewed the use of PCMs in refrigeration systems and reported their potential to improve system performance while reducing energy consumption.

Recent studies have further demonstrated the applicability of PCMs and nanoparticle-assisted thermal storage across different cooling and energy systems. Biru et al. [10] experimentally investigated a vapor-compression refrigeration system integrated with PCM and reported that a 2 mm paraffin-wax layer applied to the condenser increased the coefficient of performance by 34% while reducing the on–off cycling frequency by a factor of 1.4. Bassam et al. [11] incorporated NEPCM into a finned cooling duct beneath a photovoltaic panel and found that the use of SiC nanoparticles increased the thermal efficiency to 86.78%. Mane and Patil [12] examined paraffin-based PCM modules installed at the contact surface of a household refrigerator evaporator coil and observed improved cooling performance. In a related solar-energy application, Al-Aasam et al. [13] coupled a PCM container with a photovoltaic-thermal system and reported a maximum total efficiency of 94.31% for the configuration employing a twisted absorber tube. Recent investigations have increasingly focused on structural and material-based approaches for overcoming the low thermal conductivity of PCMs. Hierarchical and open-celled metal foams have been shown to modify heat-transfer pathways and accelerate phase-change processes, with their performance being strongly dependent on the internal porous structure and geometric characteristics [14, 15]. The simultaneous use of metal foams and nanoparticle additives has also received attention as a means of combining solid-matrix conduction with improved thermophysical properties of the storage medium [16]. Further studies have examined gradient metal-foam configurations, multiple PCMs, fins, and other heat-transfer structures to improve the charging and discharging behavior of thermal energy storage units [17–19]. In parallel, nano-enhanced PCMs have demonstrated improved thermal transport and charging/discharging characteristics in experimental thermal energy storage systems [20], while recent PCM-based storage designs have extended these concepts to application-oriented thermal management systems [21]. More recent experimental and numerical investigations have further shown that both metal-foam configuration and enclosure geometry can strongly influence phase-change behavior and overall thermal response [22, 23]. Collectively, these studies demonstrate substantial progress in material modification, porous-structure design, and geometric optimization for PCM-based thermal energy storage. However, the coupled influence of cooling-boundary arrangement, porous conductive networks, and multicomponent nanofluids remains less systematically understood, particularly during solidification in complex cold-storage geometries. A direct assessment of these mechanisms within a common configuration is therefore needed to distinguish their individual contributions and determine how their combined use affects the transient freezing process.

Against this background, particular uncertainty remains regarding the simultaneous influence of cooling-boundary arrangement, porous conductive structures, and multicomponent nanofluids on solidification. In particular, limited attention has been given to complex storage enclosures in which cooling is imposed from both the upper and lower boundaries while a porous metal foam and a ternary nanofluid are employed simultaneously. These design features influence solidification through distinct but interacting mechanisms: the dual cooling boundaries provide heat-extraction paths from different regions of the enclosure, the metal foam forms a continuous high-conductivity network within the storage medium, and the ternary nanoparticles modify its effective thermal properties. A direct comparison of these mechanisms within the same thermal storage configuration is therefore needed to determine their relative contributions to solidification performance.

Accordingly, this study investigates solidification intensification in a dual-cooled cold thermal energy storage system containing porous metal foam and an Al_2O_3 – TiO_2 –Ag/water ternary nanofluid. The transient phase-change process is numerically examined in a complex enclosure, with particular attention to temperature distribution, solid-fraction evolution, solidification-front propagation, and the time required for complete freezing. The effects of nanoparticle loading and metal-foam incorporation are evaluated both individually and in combination to distinguish their respective contributions to heat extraction. The study therefore provides a unified assessment of boundary configuration and material-based heat-transfer enhancement and clarifies which mechanism governs the acceleration of solidification under the investigated conditions. The resulting analysis is intended to support the thermal design and optimization of compact, high-response cold thermal energy storage systems.

2 Physical Model and Numerical Method

The physical model considered in this study consists of a cold thermal energy storage enclosure with two cooled boundaries located at the upper and lower surfaces, as illustrated in Figure 1. A porous metal foam is incorporated into the storage domain to provide a continuous high-conductivity pathway for heat extraction. Water is used as the base fluid, while Al_2O_3 , TiO_2 , and Ag nanoparticles are dispersed to form the ternary nanofluid. The effects of the porous matrix and ternary nanoparticles are examined both individually and in combination to determine their respective contributions to the solidification process.

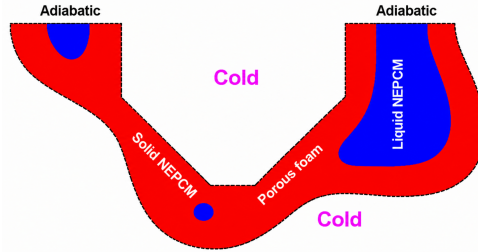


Figure 1. Porous container with complex shapes of cold surfaces

Note: NEPCM = Nanoparticle-enhanced phase change material.

Under the conditions considered, fluid motion during solidification is assumed to have a negligible influence on heat transfer. The velocity-dependent terms are therefore omitted, and the mathematical formulation is reduced to transient heat conduction coupled with the liquid-to-solid phase transition. The phase state is represented by the parameter S (dimensionless solid fraction, ranging from 0 for the fully liquid state to 1 for the fully solid state), which varies according to the local temperature T (K) relative to the prescribed phase-change interval defined by the melting temperature T_m (K) and its half-width T_0 (K). The resulting governing energy equation and phase-state relation are expressed as follows [1]:

$$(\gamma(\rho C_p)_{\text{Tnf}} + (1 - \gamma)(\rho C_p)_{\text{GI}}) \frac{dT}{dt} = (\gamma k_{\text{Tnf}} + (1 - \gamma)k_{\text{GI}}) \left(\frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial x^2} \right) + (L\rho)_{\text{Tnf}} \frac{\partial S}{\partial t} \quad (1)$$

$$\begin{cases} T > (T_m + T_0) \Rightarrow & S = 0 \\ (-T_0 + T_m) < T < (T_0 + T_m) \Rightarrow & S = (-T + 0.5T_0 + T_m) / T_0 \\ T < (T_m - T_0) \Rightarrow & S = 1 \end{cases} \quad (2)$$

where, t (s) represents time, x and y (m) denote the spatial coordinates, γ (dimensionless) is the volume fraction of the ternary nanofluid within the composite domain, and L ($\text{J} \cdot \text{kg}^{-1}$) is the latent heat of fusion of the base fluid. The effective properties of the ternary nanofluid and the gypsum layer (denoted by the subscripts Tnf and GI, respectively) are incorporated into the energy equation, including the effective density ρ ($\text{kg} \cdot \text{m}^{-3}$), the volumetric heat capacity (ρC_p) ($\text{J} \cdot \text{m}^{-3} \cdot \text{K}^{-1}$), and the effective thermal conductivity k ($\text{W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$).

The thermophysical properties of the ternary nanofluid are evaluated using a homogeneous mixture model, in which the nanoparticles are assumed to be uniformly dispersed within the base fluid. The effective density, volumetric heat capacity, thermal conductivity, and latent-heat-related properties are determined from the constituent properties and nanoparticle volume fractions. The corresponding relations are given by Eqs. (3)–(8) [2–4]:

$$\rho_{\text{Thnf}} = [(1 - \phi_1) [(1 - \phi_2) ((1 - \phi_3)(\rho_f + \rho_{s3}\phi_3)) + (\rho_{s2}\phi_2)] + \rho_{s1}\phi_1] \quad (3)$$

$$(\rho C_p)_{\text{Thnf}} = (1 - \phi_1) [(1 - \phi_2) ((1 - \phi_3)(\rho C_p)_f + (\rho C_p)_{s3}\phi_3) + (\rho C_p)_{s2}\phi_2] + (\rho C_p)_{s1}\phi_1 \quad (4)$$

$$\frac{k_{\text{nf}}}{k_f} = \frac{k_{s3} + 2k_f - 2\phi_3(k_f - k_{s3})}{k_{s3} + 2k_f + \phi_3(k_f - k_{s3})} \quad (5)$$

$$\frac{k_{\text{hnf}}}{k_{\text{nf}}} = \frac{k_{s2} + 2k_{\text{nf}} - 2\phi_2(k_{\text{nf}} - k_{s2})}{k_{s2} + 2k_{\text{nf}} + \phi_2(k_{\text{nf}} - k_{s2})} \quad (6)$$

$$\frac{k_{\text{Thnf}}}{k_{\text{hnf}}} = \frac{k_{s1} + 2k_{\text{hf}} - 2\phi_1(k_{\text{hf}} - k_{s1})}{k_{s1} + 2k_{\text{hf}} + \phi_1(k_{\text{hf}} - k_{s1})} \quad (7)$$

$$(\rho L)_{\text{Thnf}} = (\rho L)_f (1 - \phi_1) (1 - \phi_2) (1 - \phi_3) \quad (8)$$

where, the subscripts f , $s1$, $s2$, and $s3$ refer to the base fluid and the three distinct types of solid nanoparticles, respectively. The parameters ϕ_1 , ϕ_2 , and ϕ_3 (dimensionless) denote the volume fractions of the corresponding nanoparticles. The subscript Thnf denotes the ternary hybrid nanofluid. The terms k_{ntf} , k_{hntf} , and k_{Tnf} ($\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) represent the effective thermal conductivities of the nanofluid, hybrid nanofluid, and ternary nanofluid, respectively, which are calculated sequentially during the multi-step homogenization process. Additionally, the effective latent heat density $(\rho L)_{\text{Thnf}}$ ($\text{J}\cdot\text{m}^{-3}$) is evaluated based on the latent heat of the base fluid and the overall volume fraction of the suspended nanoparticles.

The governing equations were solved using FLEXPDE within a finite element framework. An adaptive mesh refinement procedure was employed to increase the spatial resolution in regions with relatively large temperature gradients and near the evolving solid–liquid interface. The transient solution was obtained using implicit time integration, with the computational discretization adjusted during the simulation to resolve the progression of the phase-change front. This numerical treatment is particularly suitable for the irregular enclosure geometry considered in the present study because local refinement can be introduced without imposing a uniformly fine mesh over the entire computational domain. The numerical formulation follows the phase-change modeling framework previously applied to thermal energy storage systems [1]. The reliability of the numerical implementation is further examined through the mesh behavior and benchmark comparison presented in the following section.

3 Results and Discussion

The numerical analysis focused on the effects of ternary nanoparticles and porous metal foam on the transient solidification behavior of the dual-cooled thermal energy storage system. Their influence was evaluated through the evolution of the computational mesh, temperature field, solid fraction, solid–liquid interface, total thermal energy, and required solidification time. Particular attention was given to distinguishing the contribution of nanoparticle-induced changes in effective thermal properties from the conductive heat-transfer pathway established by the porous metal foam.

The adaptive computational mesh used in the simulations is illustrated in Figure 2. As solidification progressed, local refinement was concentrated near the evolving phase-change front, where relatively large temperature gradients occurred. This adaptive treatment allowed the numerical resolution to follow the moving solid–liquid interface without imposing a uniformly fine mesh over the entire computational domain. The numerical implementation was further assessed by comparison with the previously published results in Sheikholeslami's work [24]. Figure 3 compares the predicted solid fraction with the corresponding benchmark values at selected times. The close agreement between the two sets of results indicates that the present numerical formulation reproduces the reference phase-change behavior with satisfactory consistency. This comparison therefore provides support for the use of the numerical framework in the subsequent analysis, although it should be regarded as a benchmark-based verification rather than experimental validation of the present storage configuration.

The transient temperature and solid-fraction distributions are presented in Figures 4–6 for the investigated configurations. During solidification, the solid region developed from the cooled boundaries and progressively advanced toward the remaining liquid zones. The regions adjacent to the horizontal adiabatic walls solidified later because heat extraction through these boundaries was restricted. The temperature contours also followed the complex enclosure geometry, reflecting the strong influence of the boundary arrangement on the local thermal field. As time increased, the average temperature of the storage medium decreased while the solid fraction increased, indicating the continuous removal of thermal energy and the advancement of the phase-change front.

Marked differences in the required freezing time were observed among the investigated cases. For the baseline configuration using water, complete solidification required approximately 322.62 s. The introduction of the ternary nanoparticles reduced the solidification time to 280.32 s, indicating that the increase in effective thermal transport provided a measurable acceleration of the phase-change process. A much stronger response was obtained when porous metal foam was incorporated into the storage domain, for which the solidification time decreased to 49.84 s. The substantially shorter freezing time can be attributed to the continuous conductive network formed by the metallic matrix, which facilitates heat extraction not only near the cooled boundaries but also in regions located farther from them. These results show that the porous metal foam provides the dominant heat-transfer enhancement under the conditions considered, while the ternary nanofluid contributes an additional but smaller improvement.

The evolution of the solid–liquid interface shown in Figure 7 further illustrates the different roles of the two enhancement mechanisms. The solidification front advanced more rapidly when thermal transport was intensified by the ternary nanoparticles and porous matrix. The effect associated with metal foam was approximately 6.3 times greater than that produced by nanoparticle addition alone. This difference is consistent with the distinct physical mechanisms involved. The ternary nanoparticles modify the effective thermophysical properties of the fluid phase, whereas the metal foam introduces a spatially continuous high-conductivity skeleton that directly connects

different regions of the storage domain. Consequently, conductive heat removal becomes considerably more effective throughout the enclosure.

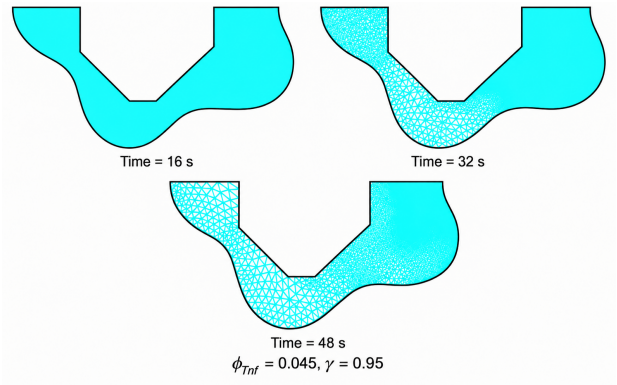


Figure 2. Mesh generation

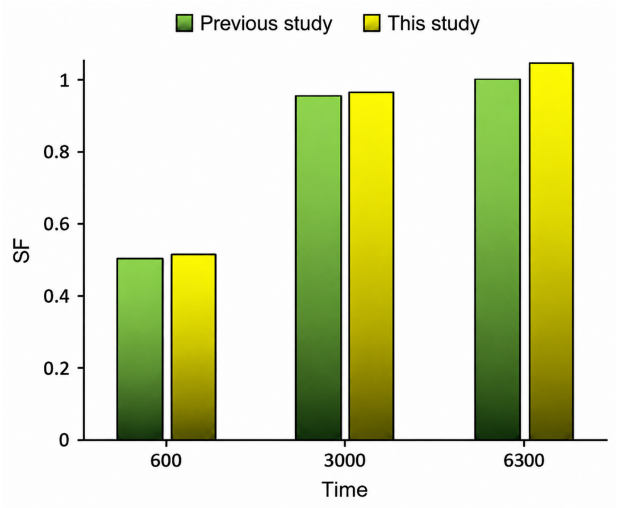


Figure 3. Comparison of the present numerical results with the benchmark data reported in Sheikholeslami's work [24]

Note: SF = solid fraction.

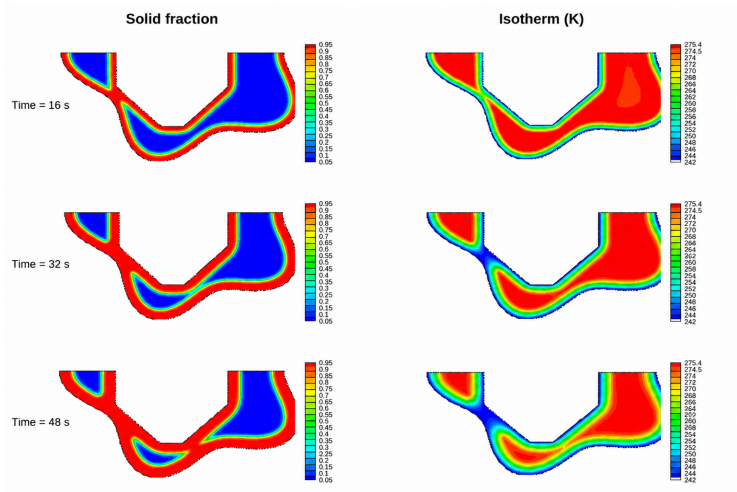


Figure 4. Evolution of the temperature field and solid fraction for $\phi_{Tnf} = 0$, $\gamma = 1$

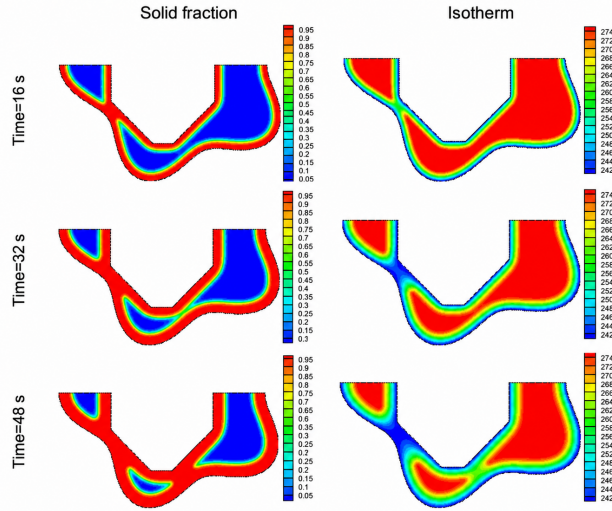


Figure 5. Evolution of the temperature field and solid fraction for $\phi_{Tnf}=0.045, \gamma=1$

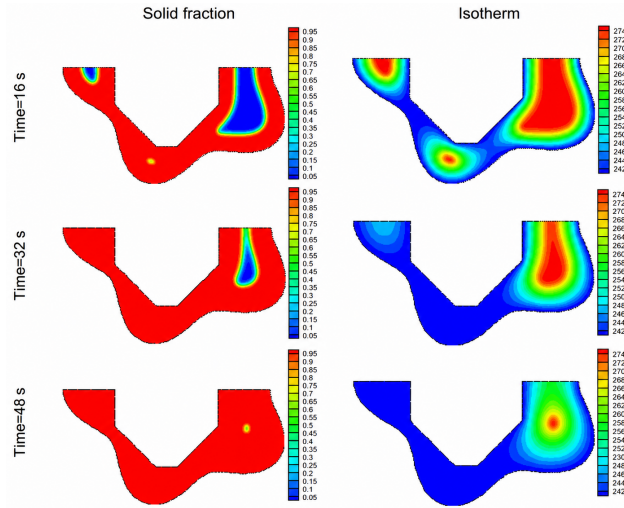


Figure 6. Evolution of the temperature field and solid fraction for $\phi_{Tnf}=0.045, \gamma=0.95$

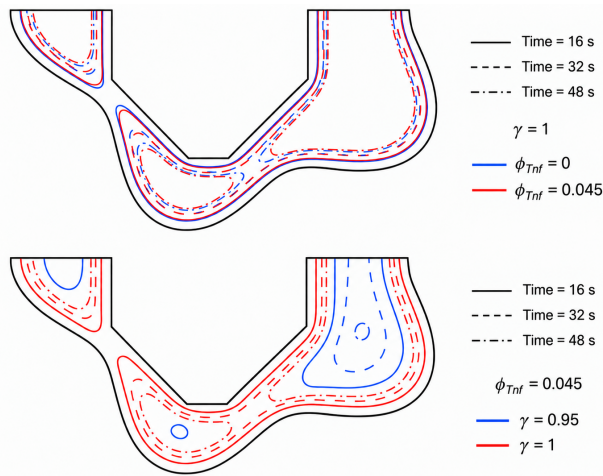


Figure 7. Evolution of the solid-liquid interface under different enhancement conditions

The temporal variations in total thermal energy, solid fraction, and average temperature are shown in Figures 8 and 9. In all cases, the solid fraction increased continuously as the phase-change process proceeded, while both the average temperature and total thermal energy decreased. These trends are consistent with progressive heat extraction from the storage medium. The presence of the porous metal foam produced a noticeably steeper thermal response, reflecting the more rapid transfer of energy from the interior of the enclosure toward the cooled boundaries. The ternary nanofluid also accelerated the thermal response, although its influence remained smaller than that of the porous structure. This comparison indicates that the overall solidification rate is governed primarily by the conductive network of the metal foam, with nanoparticle addition acting as a complementary means of improving thermal transport.

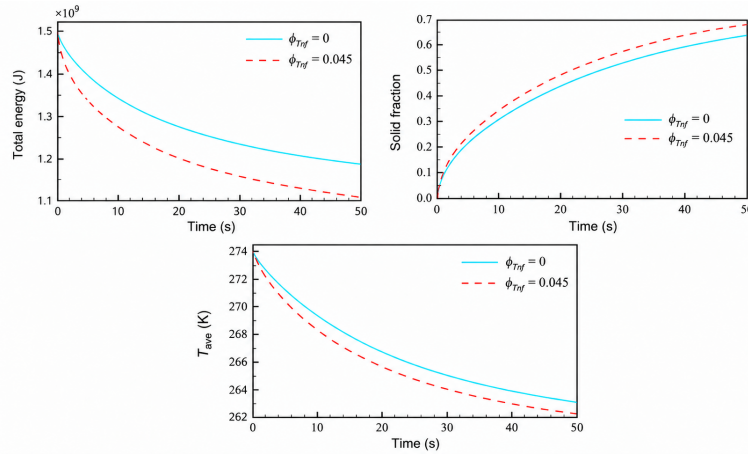


Figure 8. Effect of nanoparticle volume fraction ϕ on the transient thermal response and solidification behavior

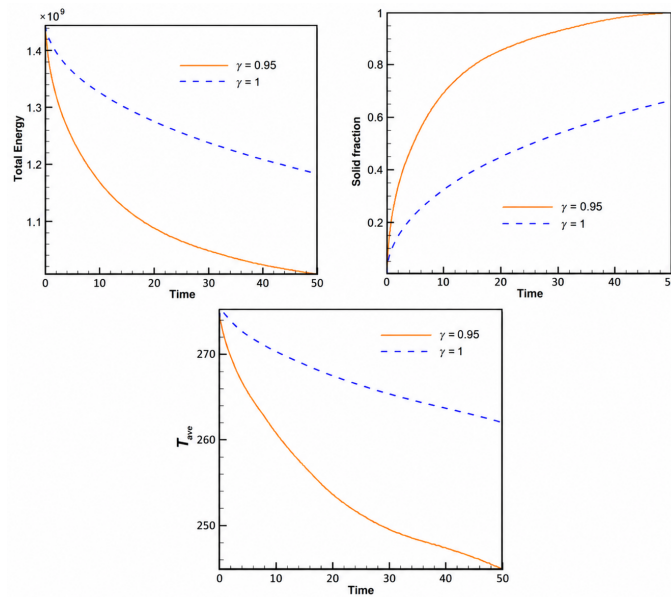


Figure 9. Effect of the metal-foam-related parameter γ on the transient thermal response and solidification behavior

Figure 10 summarizes the dependence of the required solidification time on the nanoparticle volume fraction ϕ_{Tnf} and the metal-foam-related parameter γ . Increasing ϕ_{Tnf} reduced the freezing time by approximately 13.11% relative to the case without ternary nanoparticles. This reduction is associated with the improved effective thermal transport of the Ag–TiO₂–Al₂O₃/water nanofluid. In comparison, the incorporation of porous metal foam reduced the total solidification time by approximately 82.22%, demonstrating a much stronger influence on the thermal response of the system. The greater effectiveness of the foam is associated with the formation of a continuous conductive pathway throughout the enclosure, which facilitates heat transfer in regions that are relatively distant from the cooling surfaces.

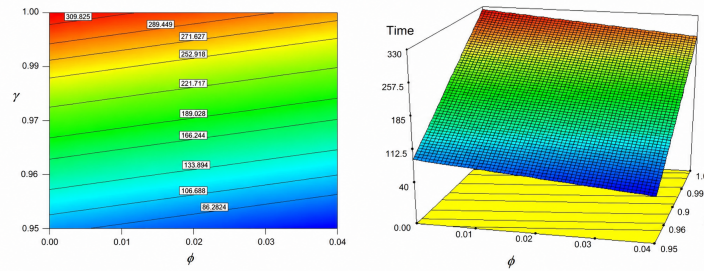


Figure 10. Effects of nanoparticle volume fraction and metal-foam parameter on the required solidification time

When the investigated enhancement mechanisms were applied together, the freezing time was reduced by approximately 84.55% relative to the baseline case. The combined configuration therefore provided the shortest solidification time among the cases considered. However, the comparison between the individual contributions shows that most of this reduction originated from the porous metal foam, while the ternary nanoparticles provided an additional improvement. From a thermal-design perspective, these results indicate that optimization of the porous conductive structure is likely to have a greater influence on solidification performance than increasing nanoparticle loading alone. The combined use of the two approaches nevertheless provides an effective strategy for achieving rapid heat extraction in compact cold thermal energy storage systems.

4 Conclusion

This study examined solidification intensification in a dual-cooled cold thermal energy storage system incorporating a porous metal foam and an $\text{Al}_2\text{O}_3\text{--TiO}_2\text{--Ag}$ /water ternary nanofluid. The numerical results showed that both enhancement approaches accelerated the phase-change process, although their contributions differed substantially.

For the baseline case using water, complete solidification required approximately 322.62 s. The addition of ternary nanoparticles reduced the freezing time to 280.32 s, corresponding to a reduction of about 13.11%. A much stronger response was obtained after introducing the porous metal foam, which reduced the solidification time by approximately 82.22%. Under the investigated combined configuration, the freezing time decreased by about 84.55%, reaching a minimum value of 49.84 s. The effect associated with the metal foam was approximately 6.3 times greater than that produced by nanoparticle addition alone.

The dominant influence of the porous metal foam is attributed to the continuous high-conductivity network formed throughout the storage domain, which facilitates heat extraction from regions located away from the cooled boundaries and accelerates the progression of the solid–liquid interface. The ternary nanofluid provides an additional contribution by improving the effective thermal transport properties of the working medium. The temperature and solid-fraction distributions further showed that the regions adjacent to the horizontal adiabatic walls were the last to complete solidification because heat removal through these boundaries was restricted.

Overall, the results indicate that the porous conductive structure is the primary factor controlling solidification acceleration under the conditions considered, while the ternary nanoparticles act as a complementary heat-transfer enhancement mechanism. The combined use of these approaches provides an effective strategy for reducing the response time of compact cold thermal energy storage systems. These findings offer useful guidance for the design and optimization of thermal storage units intended for building cooling, food preservation, and other temperature-sensitive energy applications.

Author Contributions

Conceptualization, M.H. and S.H.; methodology, M.H.; software, S.H.; validation, M.H. and S.H.; formal analysis, M.H.; investigation, M.H.; resources, S.H.; data curation, M.H. and S. H.; writing—original draft preparation, M.H.; writing—review and editing, M.H. and S. H.; visualization, S.H.; supervision, M.H.; project administration, S.H. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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